

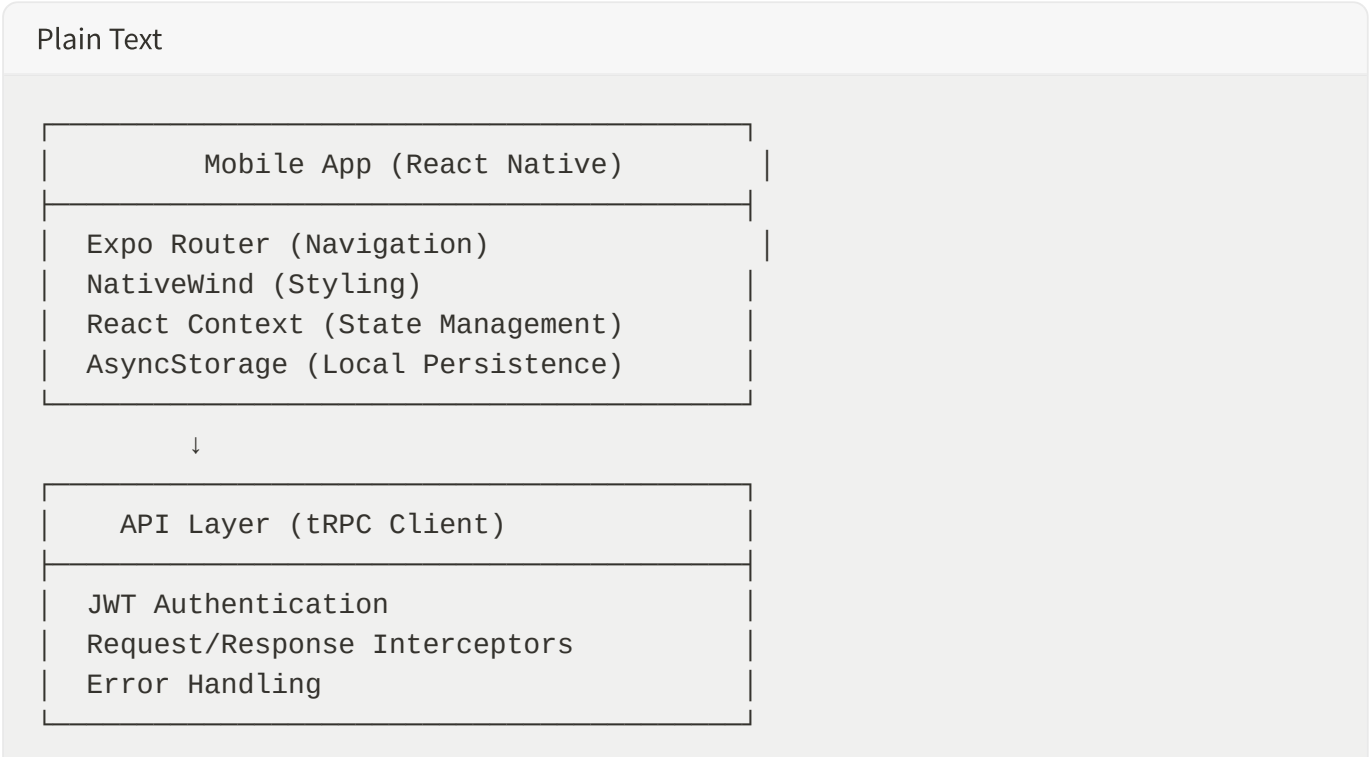
Apri Global App - Technical Documentation

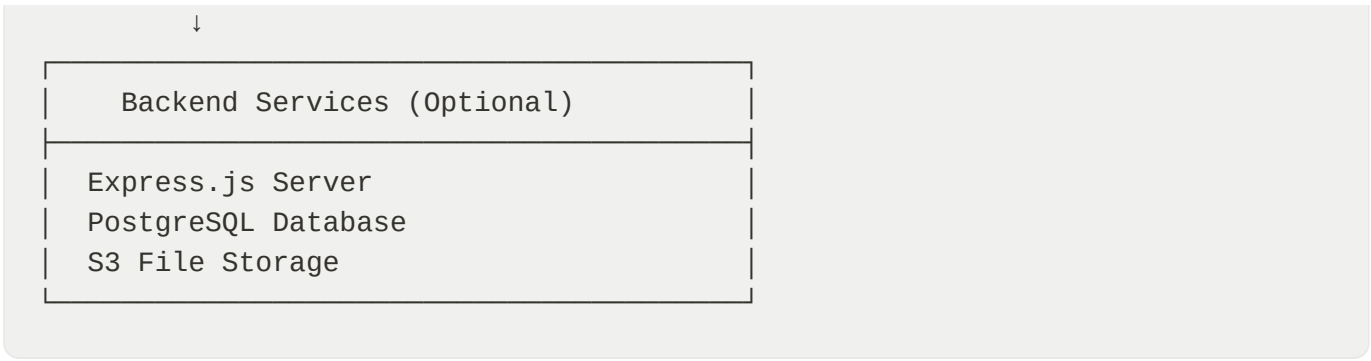
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Architecture Overview

Frontend Architecture





Key Components

Component	Purpose	Technology
Navigation	Screen routing and deep linking	Expo Router 6
State Management	Global app state	React Context + useReducer
Styling	UI styling and theming	NativeWind (Tailwind CSS)
Storage	Local data persistence	AsyncStorage / MMKV
API Client	Backend communication	tRPC + Axios
Authentication	User login and sessions	JWT tokens + OAuth 2.0

Technology Stack

Frontend

JSON

```
{  
  "runtime": "React Native 0.81.5",  
  "framework": "Expo SDK 54",  
  "language": "TypeScript 5.9",  
  "styling": "NativeWind 4.2.1 (Tailwind CSS 3.4.17)",  
  "navigation": "Expo Router 6.0.19",  
  "state_management": "React Context + useReducer",  
  "api_client": "tRPC 11.7.2 + Axios 1.13.2",  
  "animations": "React Native Reanimated 4.1.6",  
  "storage": "AsyncStorage 2.2.0 + MMKV",  
  "notifications": "Expo Notifications 0.32.15",  
}
```

```
"authentication": "Expo Secure Store 15.0.8",
"media": "Expo Image 3.0.11, Expo Video 3.0.15"
}
```

Backend (Optional)

JSON

```
{
  "runtime": "Node.js 22.13.0",
  "framework": "Express.js 4.22.1",
  "database": "PostgreSQL with Drizzle ORM 0.44.7",
  "api": "tRPC 11.7.2",
  "authentication": "JWT (jose 6.1.0) + OAuth 2.0",
  "storage": "S3-compatible (R2, MinIO)",
  "file_handling": "Multer, Busboy",
  "validation": "Zod 4.2.1",
  "testing": "Vitest 2.1.9"
}
```

Project Structure

Plain Text

```
apri-mobile-app/
├── app/
│   ├── _layout.tsx           # Root layout with providers
│   ├── oauth/                # OAuth callback handler
│   └── (tabs)/
│       ├── _layout.tsx      # Tab navigation layout
│       ├── index.tsx        # Home screen
│       ├── menu.tsx         # Menu browsing
│       ├── orders.tsx       # Order history & tracking
│       ├── rewards.tsx      # Loyalty program
│       └── account.tsx      # User profile
├── components/
│   ├── screen-container.tsx  # SafeArea wrapper
│   ├── themed-view.tsx      # Themed background view
│   └── ui/
│       ├── icon-symbol.tsx   # Icon mapping
│       └── [other UI components]
├── hooks/
│   ├── use-auth.ts          # Authentication hook
│   └── use-colors.ts         # Theme colors hook
```

```

├── use-color-scheme.ts      # Dark/light mode detection
├── lib/
│   ├── trpc.ts             # tRPC client configuration
│   ├── utils.ts            # Utility functions (cn)
│   ├── theme-provider.tsx  # Global theme context
│   └── _core/
│       ├── theme.ts        # Runtime palette builder
│       └── nativewind-pressable.ts # NativeWind Pressable fix
├── constants/
│   ├── theme.ts            # Theme exports
│   └── colors.ts           # Color definitions
├── types/
│   └── index.ts            # TypeScript type definitions
├── context/
│   └── AppContext.tsx      # Global app context
├── assets/
│   └── images/
│       ├── icon.png       # App icon
│       ├── splash-icon.png # Splash screen
│       ├── favicon.png     # Web favicon
│       └── android-icon-*.png # Android adaptive icons
├── app.config.ts           # Expo configuration
├── tailwind.config.js      # Tailwind CSS config
├── theme.config.js         # Theme color tokens
├── tsconfig.json           # TypeScript config
├── package.json            # Dependencies
└── global.css              # Global styles

server/                    # (Optional backend)
├── _core/
│   └── index.ts            # Server entry point
├── routes/
│   ├── auth.ts            # Authentication routes
│   ├── orders.ts          # Order management
│   └── [other routes]
├── db/
│   ├── schema.ts          # Database schema
│   └── migrations/        # Database migrations
└── middleware/
    ├── auth.ts            # Authentication middleware
    └── [other middleware]

```

Development Setup

Prerequisites

Bash

```
# Node.js 18+ with pnpm
node --version # v18.0.0+
pnpm --version # 9.12.0+

# Expo CLI
npm install -g expo-cli

# iOS development (macOS only)
xcode-select --install
```

Installation

Bash

```
# Clone repository
git clone [repository-url]
cd apri-mobile-app

# Install dependencies
pnpm install

# Set up environment variables
cp .env.example .env.local
# Edit .env.local with your configuration
```

Development Commands

Bash

```
# Start development server
pnpm dev

# Start with specific platform
pnpm ios # iOS simulator
pnpm android # Android emulator
pnpm web # Web browser

# Type checking
pnpm check

# Linting
pnpm lint
```

```
# Format code
pnpm format

# Run tests
pnpm test

# Build for production
pnpm build

# Generate QR code for Expo Go
pnpm qr
```

API Integration

tRPC Setup

TypeScript

```
// lib/trpc.ts
import { createTRPCReact } from '@trpc/react-query';
import type { AppRouter } from '@server';

export const trpc = createTRPCReact<AppRouter>();
```

Usage Example

TypeScript

```
// Fetching data
const { data: orders } = trpc.orders.list.useQuery();

// Mutation
const createOrder = trpc.orders.create.useMutation({
  onSuccess: (data) => {
    console.log('Order created:', data);
  },
});

// Usage
createOrder.mutate({
  items: [...],
  deliveryAddress: {...},
});
```

Authentication Flow

TypeScript

```
// 1. Login
const { mutate: login } = trpc.auth.login.useMutation();

// 2. Token storage
await SecureStore.setItemAsync('authToken', token);

// 3. API requests include token
// (Automatically handled by interceptors)

// 4. Logout
await SecureStore.deleteItemAsync('authToken');
```

Security Implementation

1. Data Encryption

TypeScript

```
// End-to-end encryption for sensitive data
import * as Crypto from 'expo-crypto';

const encryptData = async (data: string, key: string) => {
  const encrypted = await Crypto.digestStringAsync(
    Crypto.CryptoDigestAlgorithm.SHA256,
    data + key
  );
  return encrypted;
};
```

2. Secure Storage

TypeScript

```
// Store sensitive data securely
import * as SecureStore from 'expo-secure-store';

// Save token
await SecureStore.setItemAsync('authToken', token);
```

```
// Retrieve token
const token = await SecureStore.getItemAsync('authToken');

// Delete token
await SecureStore.deleteItemAsync('authToken');
```

3. Biometric Authentication

TypeScript

```
import * as LocalAuthentication from 'expo-local-authentication';

const authenticate = async () => {
  try {
    const result = await LocalAuthentication.authenticateAsync({
      disableDeviceFallback: false,
      reason: 'Authenticate to access your account',
    });
    return result.success;
  } catch (error) {
    console.error('Authentication error:', error);
  }
};
```

4. HTTPS & Certificate Pinning

TypeScript

```
// All API requests use HTTPS
// Certificate pinning configured in native layers
```

5. Input Validation

TypeScript

```
import { z } from 'zod';

const OrderSchema = z.object({
  items: z.array(z.object({
    id: z.string(),
    quantity: z.number().min(1),
  })),
  deliveryAddress: z.object({
    street: z.string(),
  })
});
```

```
    city: z.string(),
    zipCode: z.string(),
  }),
});

// Validate before sending
const validatedOrder = OrderSchema.parse(orderData);
```

Deployment Guide

iOS Deployment

Bash

```
# Build for iOS
pnpm build:ios

# Submit to App Store
eas submit --platform ios
```

Android Deployment

Bash

```
# Build for Android
pnpm build:android

# Submit to Google Play
eas submit --platform android
```

Web Deployment

Bash

```
# Build for web
pnpm build:web

# Deploy to hosting service
# (Vercel, Netlify, AWS, etc.)
```

Environment Configuration

Bash

```
# .env.local
EXPO_PUBLIC_API_URL=https://api.apri.com
EXPO_PUBLIC_APP_ENV=production
EXPO_PUBLIC_SENTRY_DSN=https://...
```

Performance Optimization

1. Code Splitting

TypeScript

```
// Dynamic imports for lazy loading
const MenuScreen = lazy(() => import('./screens/Menu'));
```

2. Image Optimization

TypeScript

```
// Use Expo Image with caching
import { Image } from 'expo-image';

<Image
  source={{ uri: 'https://...' }}
  style={{ width: 200, height: 200 }}
  cachePolicy="memory-disk"
/>
```

3. List Performance

TypeScript

```
// Use FlatList for large lists
<FlatList
  data={items}
  renderItem={({ item } ) => <ItemCard item={item} />}
  keyExtractor={(item) => item.id}
  removeClippedSubviews
  maxToRenderPerBatch={10}
/>
```

4. State Management

TypeScript

```
// Use selectors to prevent unnecessary re-renders  
const userName = useAppContext(state => state.user.name);
```

Testing Strategy

Unit Tests

Bash

```
pnpm test
```

Example Test

TypeScript

```
// __tests__/utils.test.ts  
import { describe, it, expect } from 'vitest';  
import { formatPrice } from '@lib/utils';  
  
describe('formatPrice', () => {  
  it('should format price correctly', () => {  
    expect(formatPrice(99.99)).toBe('$99.99');  
  });  
});
```

E2E Testing (Optional)

Bash

```
# Using Detox for E2E testing  
detox test e2e/firstTest.e2e.js --configuration ios.sim.debug
```

Monitoring & Analytics

Error Tracking (Sentry)

TypeScript

```
import * as Sentry from 'sentry-expo';

Sentry.init({
  dsn: process.env.EXPO_PUBLIC_SENTRY_DSN,
  environment: process.env.EXPO_PUBLIC_APP_ENV,
});
```

Performance Monitoring

TypeScript

```
// Track custom metrics
Sentry.captureMessage('Order placed successfully', 'info');

// Track errors
try {
  // code
} catch (error) {
  Sentry.captureException(error);
}
```

User Analytics

TypeScript

```
// Track user events
const trackEvent = (eventName: string, properties?: object) => {
  // Send to analytics service
  console.log(`Event: ${eventName}`, properties);
};

// Usage
trackEvent('order_completed', { orderId: '123', total: 99.99 });
```

Troubleshooting

Common Issues

Issue	Solution
Metro bundler crash	Clear cache: <code>pnpm start --reset-cache</code>
Dependency conflicts	Reinstall: <code>rm -rf node_modules && pnpm install</code>
iOS build failure	Clear Xcode cache: <code>xcode-select --reset</code>
Android build failure	Clear Gradle cache: <code>cd android && ./gradlew clean</code>

Debug Mode

Bash

```
# Enable debug logging
DEBUG=* pnpm dev

# React Native debugger
pnpm start --dev-client
```

Support & Resources

- Expo Documentation: <https://docs.expo.dev>
- React Native Docs: <https://reactnative.dev>
- tRPC Documentation: <https://trpc.io>
- Tailwind CSS: <https://tailwindcss.com>
- TypeScript: <https://www.typescriptlang.org>

Version Information

- Expo SDK: 54
- React Native: 0.81.5
- TypeScript: 5.9
- Node.js: 18+
- pnpm: 9.12.0

Last Updated: March 11, 2026

Status: Production Ready